

3 Message Specifications for UDP Multicast

3.1 Protocol

3.1.1 Message Format

- The protocol for multicast distribution is UDP/IP.
- The packet distributed consists of an Ethernet header, IP header, UDP header and tag data.
- The tag data consists of a packet header from "[3.2.1 Packet Header](#)" and a tag or tags from "[3.2.2 Tag Types](#)".

Name	Offset	Length	Value	Notes	Example
Packet Header	0	26	Char/Bn		
Tag Length (1)	26	1	Bn	Set as length of first tag.	
Tag Data (1)	27	variable	Char/Bn		
...					
Tag Length (n)	variable	1	Bn	Set as length of n-th tag.	
Tag Data (n)	variable	variable	Char/Bn		



The number of tags in a packet

-The number of tags in a packet is determined by the maximum amount of data that can be delivered. In the current tag specifications, we expect the maximum to be 119 tags (T tag and 118 D tags). Information in one tag is never divided into different packets.

- The values used in tag data are as follows.

Value	Description	Main examples
Char	Indicates a character code. ASCII; left-aligned.	Tag Type, Issue Code, Side, etc.
Bn	Indicates a numerical value encoded by big-endian binary. Unsigned integer type.	All numerical items except Price
Bn (Price)	Indicates an unsigned long long type encoded by big-endian binary. The last four digits are decimal fractions.	Price

3.1.2 Tag Blocking

- When multiple tags are generated at the same time, these tags are blocked together into a packet.
- Since there is an upper limit on the number of tags that can be blocked, if the number of tags generated is over this limit, the packet will be split.
- When a packet is split, whether there is another packet or not can be identified by "Utility Flag".
- Since tags are blocked independently for each issue and each process, data for different issues are never mixed and data for different process, even for the same issue, are never mixed in the same packet.

3.2 Tag Data Types

3.2.1 Packet Header

Name	Offset	Length	Value	Notes	Example
Multicast Group Number	0	1	Bn	- Set to the Multicast Group Number.	52
Number of System Reboots	1	1	Bn	- Set to the number of times the system has been rebooted.	0
Sequence Number	2	4	Bn	- Assigned from 1 on a multicast group basis. In principle, the numbers are consecutive, but if a single DF server failure or the system is rebooted, they are assigned from 1 again. - The combination with Number of System Reboots is unique through each business day.	10030501
Issue Code	6	12	Char	Set to the Securities Identification Code as assigned by the Securities Identification Code Committee. For FLEX MBO, Issue Code is set with left-align. (For FLEX Standard, with right-align.) However, in the case of a Reset tag (<u>R tag</u>) and a Communication Control (<u>L tag</u>), this will be set to a space.	8697
Update Number	18	4	Bn	- Assigned from 1 on a per-issue basis for each business day. In principle, the numbers are consecutive, but if the system is rebooted, they are assigned from 1 again. - Even when a packet is split, the same value will be assigned to information which are generated by one transaction of a Trading Server. - The combination with Number of System Reboots is unique through each business day. However, in the case of a Reset tag (<u>R tag</u>) and a Communication Control (<u>L tag</u>), this will be set to '0'.	6000207
Packet Number	22	1	Bn	- Always set to '0'.	0
Total Number of Packets	23	1	Bn	- Always set to '0'.	0
Utility Flag	24	1	Bn	(For UDP multicast, except for Automatic sending of snapshots) When tag data which are generated by one transaction of a Trading Server cannot be distributed in a single packet, whether there is another packet following or not can be identified by this flag. When there is another packet: set to '1'. Other than the above: set to '0'.	0
Message Count	25	1	Bn	Set to the number of tags in the packet.	20

Note: Please refer to '[5.2 Notes for Receiving Snapshot Information](#)' for the values set in packet headers for snapshot information.

3.2.2 Tag Types

- Tags supporting the MBO service are as follows.

Type	Name	Length	Notes
<u>T tag</u>	Seconds Timestamp	5	- Provides information about the time of processing on the Trading Server to the nearest second. - The time to the nearest microsecond is set in the time field of each tag. - Always delivered at the time the information is generated(Except in the case of delivering a Reset tag (<u>R tag</u>) and a Communication Control (<u>L tag</u>)). - Only one tag is delivered per process. If the packet is split, it is stored at the top of the first packet.
<u>O tag</u>	Trading Status	18	- Provides information about the status of trading on a per-issue basis. - Delivered when there is a change in the tag.
<u>K tag</u>	Execution Summary	46	- Provides summary information about execution. - Delivered when an execution occurs.
<u>A tag</u>	Add Order	26	- Provides information about book registration. - Delivered when a new order or modification order is registered.
<u>E tag</u>	Order Executed	20	- Provides information about registered orders in case of Zaraba execution. - Delivered when an execution occurs in Zaraba.
<u>C tag</u>	Order Executed with Price	29	- Provides information about registered orders in case of Itayose execution. - Delivered when an execution occurs by the Itayose method or an execution occurs while an SQ or STQ is displayed.
<u>D tag</u>	Order Delete	11	- Provides information about order deletion from the order book. - Delivered when an order is deleted via a cancel order or a modification order or when a registered order expires.
<u>R tag</u>	Reset	2	- A tag requesting the resetting (i.e., cancellation) of all distributed information. - When the user receives this tag, the user must clear all valid A tags.
<u>L tag</u>	Communication Control	3	- Provides information about communication control. - Delivered when communication starts and ends, and in between at minute intervals.
II tag	Issue Information	125	Please refer to 'Common Items' and 'Issue Basic Information' in the FLEX specifications.
BP tag	Base Price Information	68	Please refer to 'Common Items' and 'Issue Basic Information' in the FLEX specifications.
MG tag	Multicast Group Number Information	16	Please refer to 'Common Items' and 'Issue Basic Information' in the FLEX specifications.

3.2.3 Tag Formats

(1) Seconds Timestamp – T tag

T tag					
Name	Offset	Length	Value	Notes	Example
Message Type	0	1	Char	Set to 'T'.	T
Time (Seconds)	1	4	Bn	Set to UNIX time (seconds since 1970-01-01 00:00:00UTC).	1704068110

(2) Trading Status – O tag

O tag					
Name	Offset	Length	Value	Notes	Example
Message Type	0	1	Char	Set to 'O'.	O
Time (Microseconds)	1	4	Bn	Indicates the time to the nearest microsecond.	323234
Market Status	5	1	Bn	Indicates market status. 11: Morning session order acceptance time 21: Afternoon session order acceptance time 12: Morning session trading hours 22: Afternoon session trading hours - 23: Afternoon session pre-closing 19: End of morning session 29: End of afternoon session	12
Status Flag	6	2	Char	Indicates trading status. A0: Trading halted (accepting orders) A1: Trading halt lifted B0: Itayose B1: Itayose ended C0: Trading suspended C1: Trading suspension lifted D0: Trading halted (not accepting orders) Space: Other than the above	Space Space
Short Selling Status	8	1	Bn	Indicates application of the Short Selling Regulation. 0: Short Selling Regulation is NOT applied. 1: Short Selling Regulation is applied.	1
Pricing Method	9	1	Bn	Indicates method of pricing or status of quote display after the tag is delivered. 1: Itayose method 2: Zaraba method 3: Sell Special Quote displayed 4: Buy Special Quote displayed 5: Sell Sequential Trade Quote displayed	4

O tag					
Name	Offset	Length	Value	Notes	Example
				6: Buy Sequential Trade Quote displayed	
				0: Other than the above (e.g., at the timing of session end)	
Book Center Price	10	8	Bn (Price)	Set to the book center price while the Itayose method is in place or while a quote is displayed.	1050000

(3) Execution Summary – K tag

K tag					
Name	Offset	Length	Value	Notes	Example
Message Type	0	1	Char	Set to 'K'.	K
Time (Microseconds)	1	4	Bn	Set to the time to the nearest microsecond.	323234
Triggered Side	5	1	Char	Indicates the side whose order triggered the execution.	B
				S: Sell order	
				Space: Itayose execution	
				B: Buy order	
Total Volume	6	6	Bn	Set to the total volume of executed orders (number of shares).	10000
Total Invalidation	12	6	Bn	Set to the total volume of invalidated orders (number of shares).	200
Last Price	18	8	Bn (Price)	Set to the last price at the time the execution was processed.	1000000
Match ID	26	4	Bn	An identifier for each issue is assigned on an execution basis.	601257
				It is unique and sequential through each business day.	
Best Offer	30	8	Bn (Price)	Set to the best offer at the time the execution processing finishes.	1020000
				At market close, it is set to '0'.	
Best Bid	38	8	Bn (Price)	Set to the best bid at the time the execution processing finishes.	980000
				At market close, it is set to '0'.	

(4) Add order – A tag

A tag					
Name	Offset	Length	Value	Notes	Example
Message Type	0	1	Char	Set to 'A'.	A
Time (Microseconds)	1	4	Bn	Set to the time to the nearest microsecond.	323234
Order ID	5	4	Bn	An identifier for each issue is assigned on an order basis.	1001054
				It is unique and sequential through each business day.	
Side	9	1	Char	Indicates buy or sell.	S
				S: Sell order B: Buy order	
Quantity	10	6	Bn	Set to the order quantity registered on the order book (number of shares).	2000
Price	16	8	Bn (Price)	Set to the price registered on the order book.	1050000
				For market orders, this will be set to the maximum value (64bit).	

A tag					
Name	Offset	Length	Value	Notes	Example
Order Condition	24	1	Bn	Indicates the order condition. 0: Non-conditional 2: On-open 4: On-close 6: Funari	0
Modification Flag	25	1	Bn	When the A tag is generated by a modification order, this indicates whether there is a change of time priority or not. 1: Modification order that does NOT change its time priority (reducing the number of ordered shares, etc.) 0: Cancel order or modification order that changes its time priority	1

(5) Order Executed – E tag

E tag					
Name	Offset	Length	Value	Notes	Example
Message Type	0	1	Char	Set to 'E'.	E
Time (Microseconds)	1	4	Bn	Set to the time to the nearest microsecond.	323234
Order ID	5	4	Bn	Set to the Order ID that was assigned to the executed order. When a partial execution occurs, the same Order ID may be set multiple times.	1001054
Side	9	1	Char	Indicates sell or buy. S: Sell order B: Buy order	B
Volume	10	6	Bn	Set to the executed volume (number of shares).	100
Match ID	16	4	Bn	An identifier for each issue is assigned on an execution basis. It is unique and sequential through each business day.	601257

(6) Order Executed with Price – C tag

C tag					
Name	Offset	Length	Value	Notes	Example
Message Type	0	1	Char	Set to 'C'.	C
Time (Microseconds)	1	4	Bn	Set to the time to the nearest microsecond.	323234
Order ID	5	4	Bn	Set to the Order ID that was assigned to the executed order. When a partial execution occurs, the same Order ID may be set multiple times.	1001054
Side	9	1	Char	Indicates sell or buy. S: Sell order B: Buy order	B
Volume	10	6	Bn	Set to the executed volume (number of shares).	100
Match ID	16	4	Bn	An identifier for each issue is assigned on an execution basis. It is unique and sequential through each business day.	601257
Execution Price	20	8	Bn (Price)	Set to the execution price.	1080000

C tag					
Name	Offset	Length	Value	Notes	Example
Adopted Pricing Method	28	1	Bn	Indicates the pricing method in place when the execution occurred. 1: Itayose method 2: Quote displayed	1

(7) Order Delete – D tag

D tag					
Name	Offset	Length	Value	Notes	Example
Message Type	0	1	Char	Set to 'D'.	D
Time (Microseconds)	1	4	Bn	Set to the time to the nearest microsecond.	323234
Order ID	5	4	Bn	Set to the Order ID that was assigned to the cancelled order.	1001054
Side	9	1	Char	Indicates sell or buy. S: Sell order B: Buy order	S
Modification Flag	10	1	Bn	When the D tag was generated by a modification order, this indicates whether there is a change of time priority or not. 1: Modification order that does NOT change its time priority (reducing the number of ordered shares, etc.) 0: Cancel order, modification order that changes its time priority, or expired order	1

(8) Reset - R tag

R tag					
Name	Offset	Length	Value	Notes	Example
Message Type	0	1	Char	Set to 'R'.	R
Start-End Flag	1	1	Bn	When TSE needs to reset all distributed data, this indicates the start or end of the process. 1: Start (set when dissemination of the reset information is started) When the user receives this tag, the user must clear all valid A tags received prior to that point. 2: End (set when dissemination of the reset information is completed)	1

(9) Control - L tag

L tag					
Name	Offset	Length	Value	Notes	Example
Message Type	0	1	Char	Set to 'L'.	L
Test Mode Flag	1	1	Bn	Indicates production or test. 1: Production 2: Test	1
Start-End Flag	2	1	Bn	Indicates the start or end of communication. 1: Communication Start 2: Communication End 0: Health Check control	1

3.2.4 Tag Priorities

- When multiple tags are generated by the same process, tag priorities are, in principle, as follows.

Priority	Tag Types	Notes
1	T tag	- Always delivered at the time information is generated.
2	O tag	- Delivered when there is a change in the tag. If there is no change, it is omitted.
3	K tag	- Delivered when an execution occurs. If there is no execution, it is omitted.
4	D tag	- Provides information about order deletion through modification orders or cancel orders, etc.
5	E tag/ C tag, D tag	- Provides information about execution or invalidation. On the same price level, execution (E tag or C tag) comes before invalidation (D tag). When information occurs on multiple price levels, for example when a buying-up occurs, the order of distribution is E tag, D tag, E tag, and so on.
6	A tag	- Provides information about registered order volumes stemming from new orders and modification orders.

**Priorities among C tags**

-In case of delivering C tags due to normal Itayose execution occurrence, the priorities among C tags are as follows.

(1st) Orders executed at the price at which there are remaining orders(either buy or sell),

(2nd) Buy orders at the price from lower to market buy order, (3rd) Sell orders at the price from higher to market sell order.

3.3 Appendix

3.3.1 Multicast Routing Maintenance Message

- A Multicast Routing Maintenance message is delivered for each multicast group.
- The message doesn't have a packet header, and 1-byte spaces '0x20' are set in the data area.
- The purpose of the message is to maintain and establish a multicast distribution tree, and to prevent message duplication before delivering the control messages (Communication Start and Health Check control), by sending it to the same multicast address as other messages.
- Although the message is distributed from the exchange-side system to users, it is not necessary for the user-side application to capture the message, as long as it reaches the user-side system.

3.3.2 Special Examples to Consider for User-side Applications

(1) On-close orders/Funari orders at and after the start of Pre-closing

- In a session with a closing auction, On-close orders are brought onto the order book as valid orders after Pre-closing starts.
- In a session with a closing auction, Funari orders are brought onto the order book, not as limit orders but as market orders, after Pre-closing starts.
- At the time Pre-closing starts, information about On-close and Funari orders being brought onto the order book as above is not distributed.
- Also, On-close orders and Funari orders are sometimes sent during Pre-closing.
- If a user constructs an order book or calculates indicative closing prices based on information from the MBO service, it is necessary to consider the above examples on the user-side system.

(2) A value set in 'Utility Flag' tag in packet header in a UDP multicast

- In opening auctions or other Itayose auctions, a large amount of execution information is generated.
- When tag data which are generated by one matching process cannot be distributed in a single packet, the packet is split.
- Also, in order to deliver an Execution Summary message without delay, a packet that contains a K tag is sometimes distributed prior to other packets.
- In the above situations, by setting the Utility Flag value to '1: When there is another packet', it is possible to identify whether the information was generated by the same matching process or not.
- The same method is used to deliver messages in the case of Zaraba execution where there is a large number of orders executed.
- 'Packet Number' and 'Total Number of Packets' are set to '0' at the time of arrowhead4.0 operation.

(Distribution example) Itayose execution in an opening auction

Packet No.	Total No. of Packets	Utility Flag	Example of Tags	Notes
0	0	1	T O K C C ... C	In order to deliver the K tag without delay, the first packet is delivered before the other packets. Since the Utility Flag is set to '1', there will be more messages that were generated by the same matching process.
0	0	1	C C C ... C C	Since the Utility Flag is set to '1', there will be more messages that were generated by the same matching process.
:	:	:		
0	0	1	C C C ... C C	Since the Utility Flag is set to '1', there will be more messages that were generated by the same matching process.
0	0	1	C C C ... C D	Since the Utility Flag is set to '1', there will be more messages that were

Packet No.	Total No. of Packets	Utility Flag	Example of Tags	Notes
				generated by the same matching process.
0	0	0	D D D . . . D D	The last packet

(3) **Example of where 'Price' in an A tag is replaced, e.g., when the initial price of a newly listed issue is determined**

- Before the initial price of a newly listed issue is determined, the price limit range is not prescribed since there is no base price. When an initial price is determined during a session, the upper/lower limit prices will then be determined based on that initial price. Given this, On-close buy orders that had been placed at a price higher than the new upper limit price and On-close sell orders that had been placed at a price lower than the new lower limit price will be treated as orders placed at the upper/lower limit price.
- When a base price is changed after order acceptance starts, buy orders that had been placed at a price higher than the new upper limit price and sell orders that had been placed at a price lower than the new lower limit price will be treated as orders placed at the upper/lower limit price.
- Since A tags will not be redistributed in the above situations, it is necessary to consider this on the user-side system.